

iREPAIR



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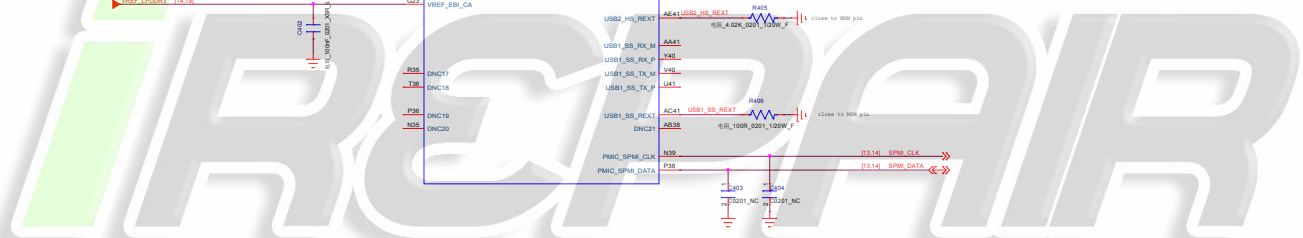


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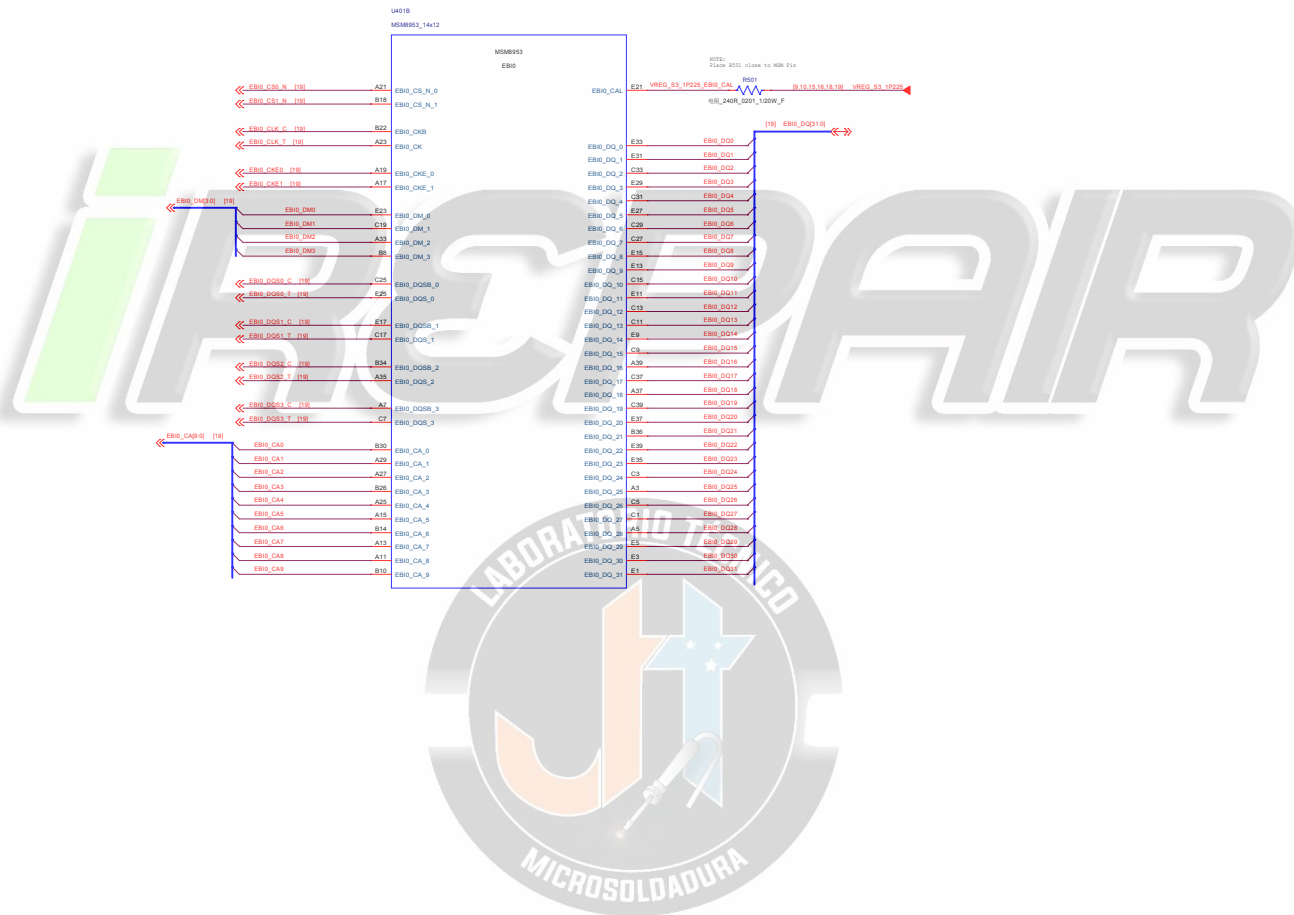
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MSM8953
Control

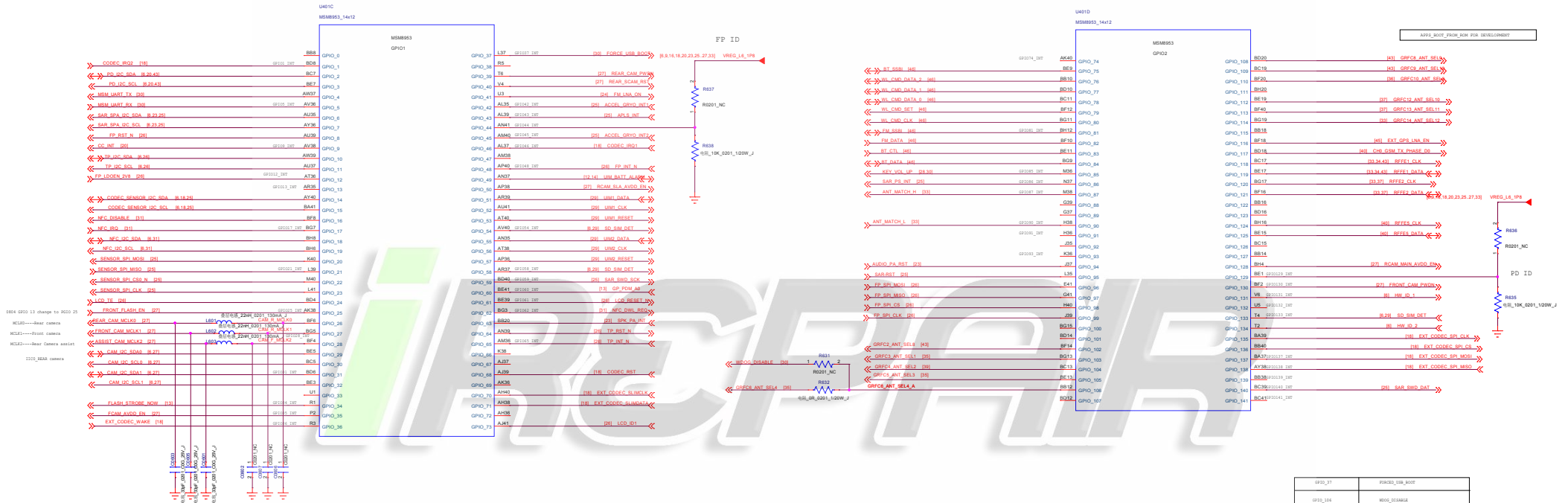
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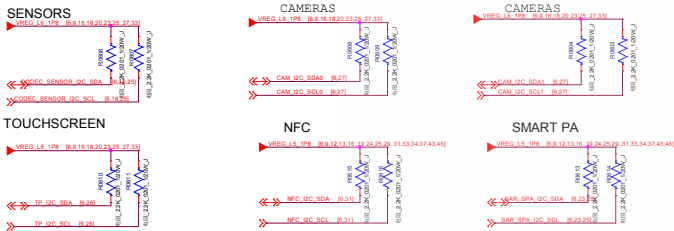
MSM8953
EBI

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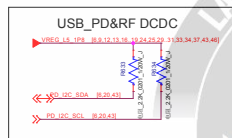
NOTE:
Asterisks (*) indicate modem power management (MPM) wake-up pins



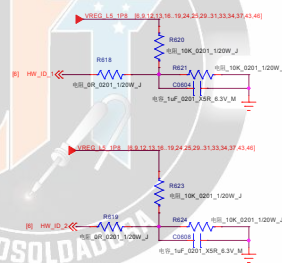
I2C PULL-UP



NOTE:
Ensure SW sets these GPIOs (Sensor, CTP and Camera I2C bus) to input pull down when the peripherals are powered off to eliminate leakage.



HW Board ID

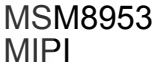


MSM8953 GPIO

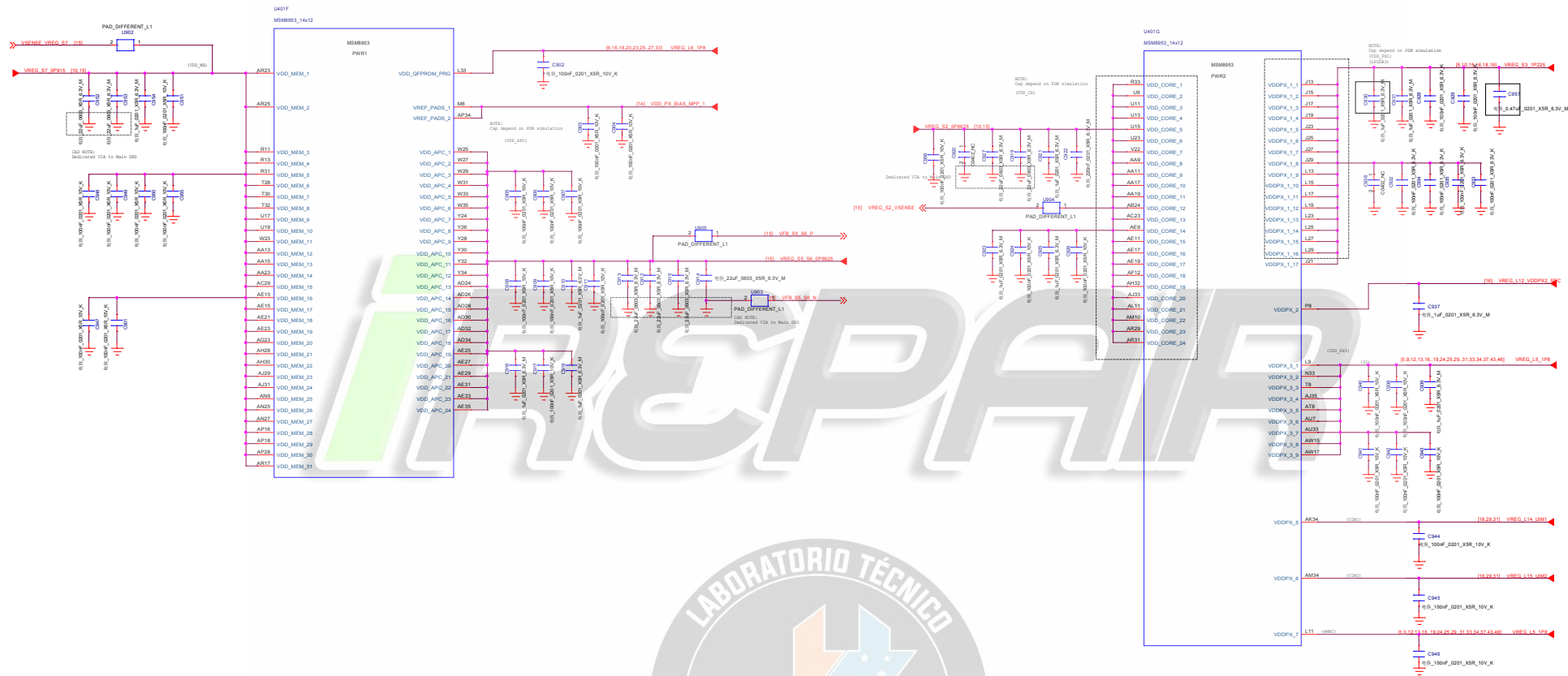
GPIO_37	GPIO37_000
GPIO_104	GPIO104_000
GPIO_108	GPIO108_000

Default Boot Config (GPIOs) is GPIO108

Pin	GPIO
Pin	GPIO
Pin	GPIO

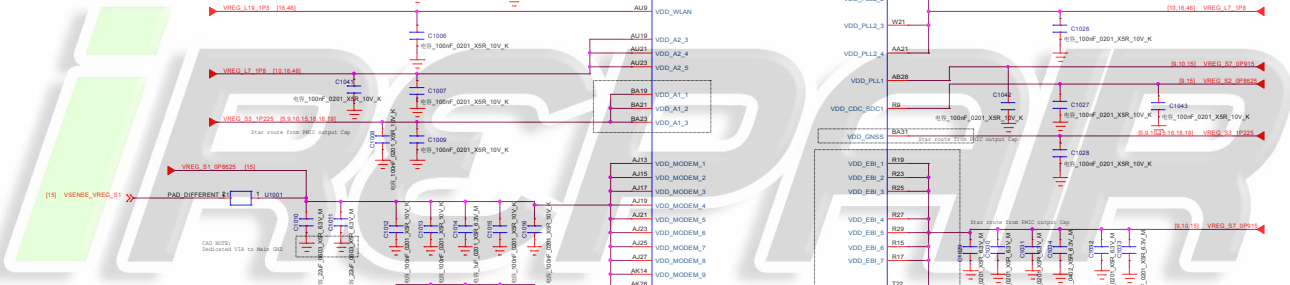


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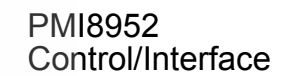


MSM8953 POWER1

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MSM8953
POWER2

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VREG_S3_1P225作为输入电压太低

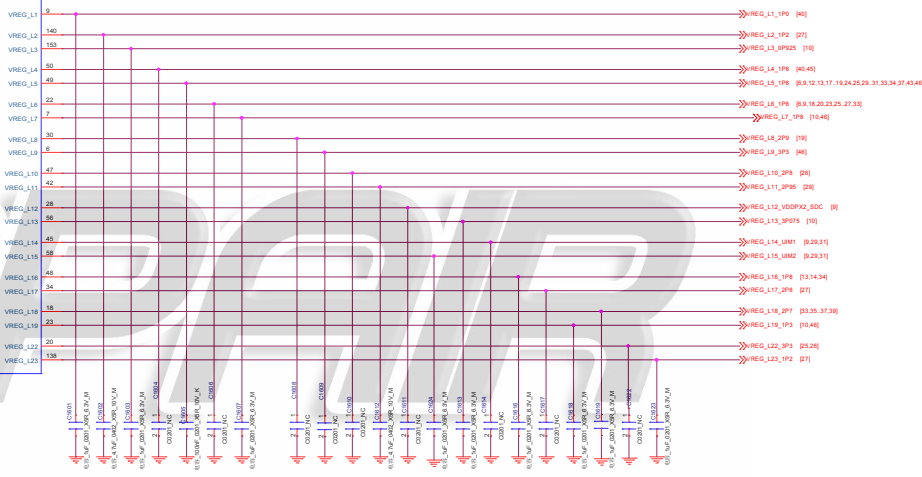
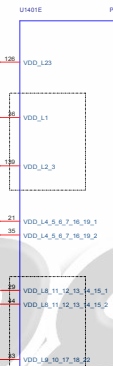
VREG_S3_1P225 [0.9,10,15,16,18,19]

VREG_S3_1P225 [0.9,10,15,16,18,19]

VREG_S4_2P05 [14,15,17]

VPH_PWR [12,15,17,18,20,23,25,27,31,33,34,43,44]

Nonlinear routing to each VIO group



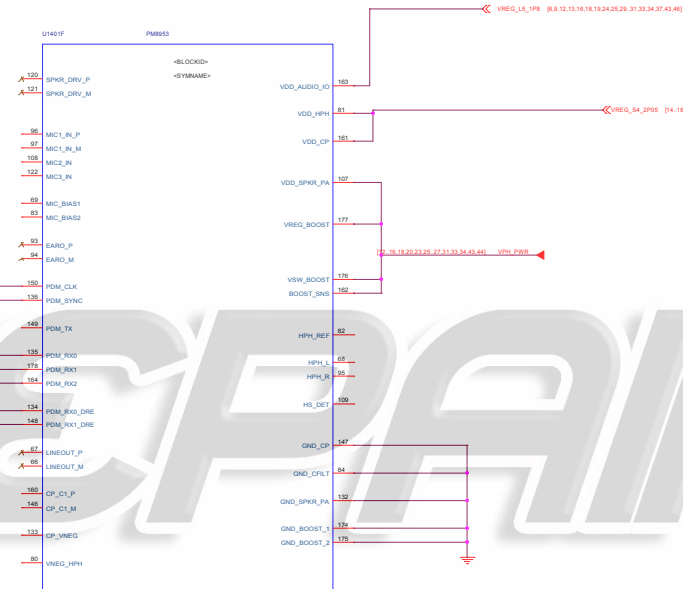
L1	PMOS LDO	1.0	1.0	0.375-1.5375	800	N	REFs	
L2	PMOS LDO	1.708	1.708	0.375-1.5375	1200	N	Camera + Display	
L3	PMOS LDO	0.925	0.925	0.375-1.5375	800	Y	MEM DSI PLL, USB	
L4	PMOS LDO	1.800	1.800	1.750-3.3375	450	N	REFs and GPS vRAM	
L5	PMOS LDO	1.800	1.800	1.750-3.3375	600	Y	Most digital ICs, MEM and group 3 and 7, LPDDR, and eMMC	
L6	PMOS LDO	1.800	1.800	1.750-3.3375	700	N	MEM, CAMERA, camera, touchscreen, display, and sensor	
L7	PMOS LDO	1.800	1.800	1.750-3.3375	500	Y	MEM, display, USB, and PLLs, WCN, KIO, and PMU based clock driver	
L8	PMOS LDO	2.900	2.900	1.750-3.3375	800	Y	eMMC	
L9	PMOS LDO	Vout = 3.3 V for VBAT = 3.075 V, Vout = 3.3 V for VBAT = 3.075 V		3.000-3.300	1.750-3.3375	800	N	WCN

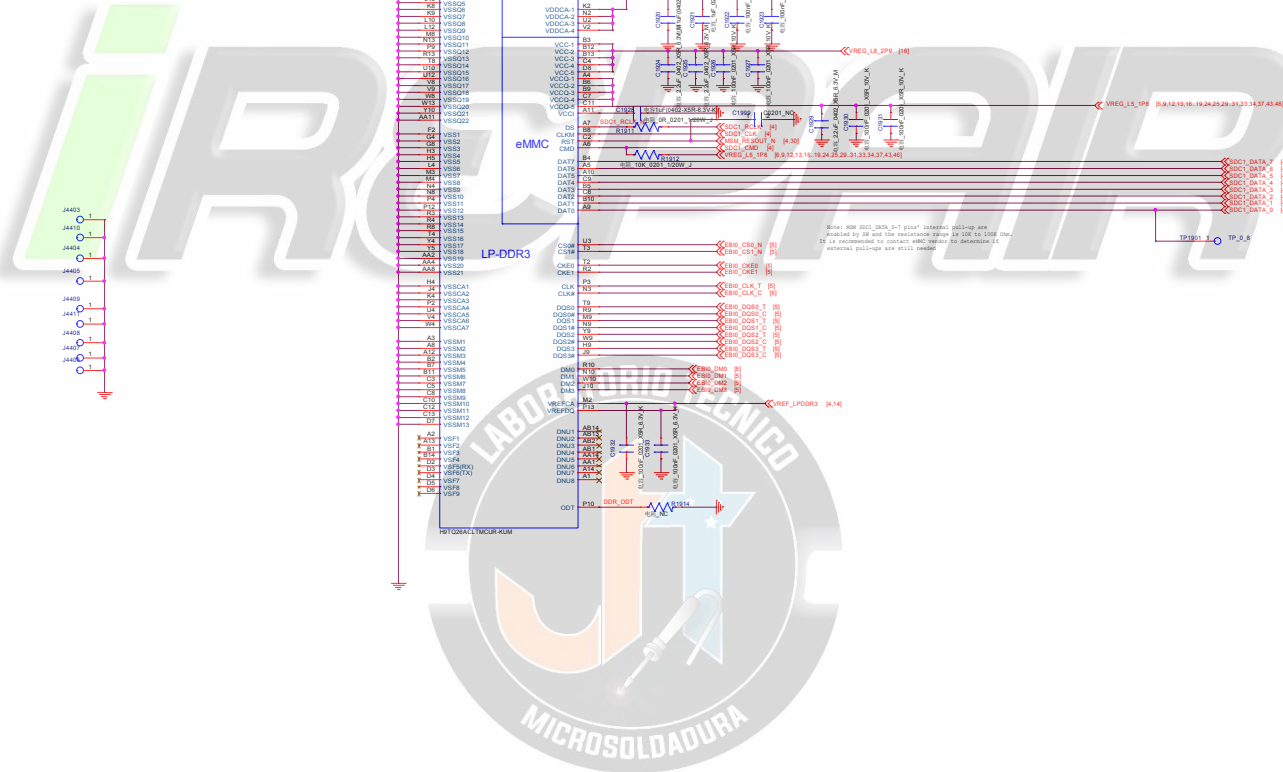
Partnumber	Circuit type	Default V (V)	Specified range (V)	Programmable range (V)	Rated current (mA)	Default ON	Expected use (MEMS)
L10	PMOS LDO	3.0	3.0	1.750-3.3375	100	N	Sensor and touchscreen
L11	PMOS LDO	2.900	2.900	1.750-3.3375	800	Y	Mass SD
L12	PMOS LDO	2.900	1.800-2.900	1.750-3.3375	50	Y	MEM per group 2 and SOC2
L13	PMOS LDO	3.125	3.125	1.750-3.3375	100	Y	MEM USB type C and PMIC and external codec
L14	PMOS LDO	1.800	1.800	1.750-3.3375	50	N	MEM per group 6: dual voltage USB1, and NFC
L15	PMOS LDO	1.800	1.800	1.750-3.3375	50	N	MEM per group 6: dual voltage USB2
L16	PMOS LDO	1.800	1.800	1.750-3.3375	5	N	PMIC REFs
L17	PMOS LDO	2.900	2.900	1.750-3.3375	300	N	Camera and display
L18	PMOS LDO	3.700	3.700	1.750-3.3375	100	N	Q1, RF, and other
L19	PMOS LDO	1.300	1.300	0.375-1.5375	800	N	MEM analog, WCN, and WGR
L20	Low-noise LDO	1.7A	1.7A	1.7A-3.3375	5	Y	PMIC SOC mode
L21	Low-noise LDO	1.7A	1.7A	1.7A-3.3375	5	N	PMIC RF mode
L22	PMOS LDO	2.800	2.800	1.750-3.3375	150	N	Camera - analog
L23	PMOS LDO	1.10	1.10	0.375-1.5375	800	N	Camera - digital



PM8953
LDO

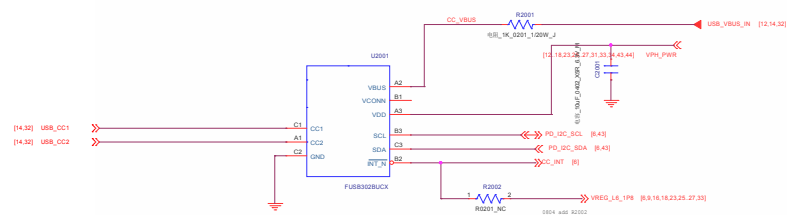
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MEMORY(LPDDR3+EMMC V5

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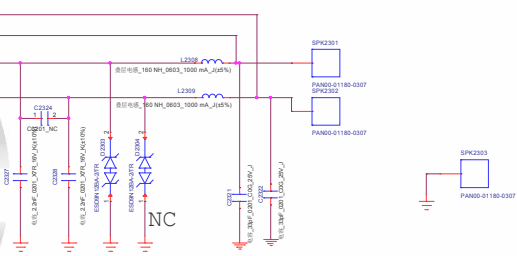
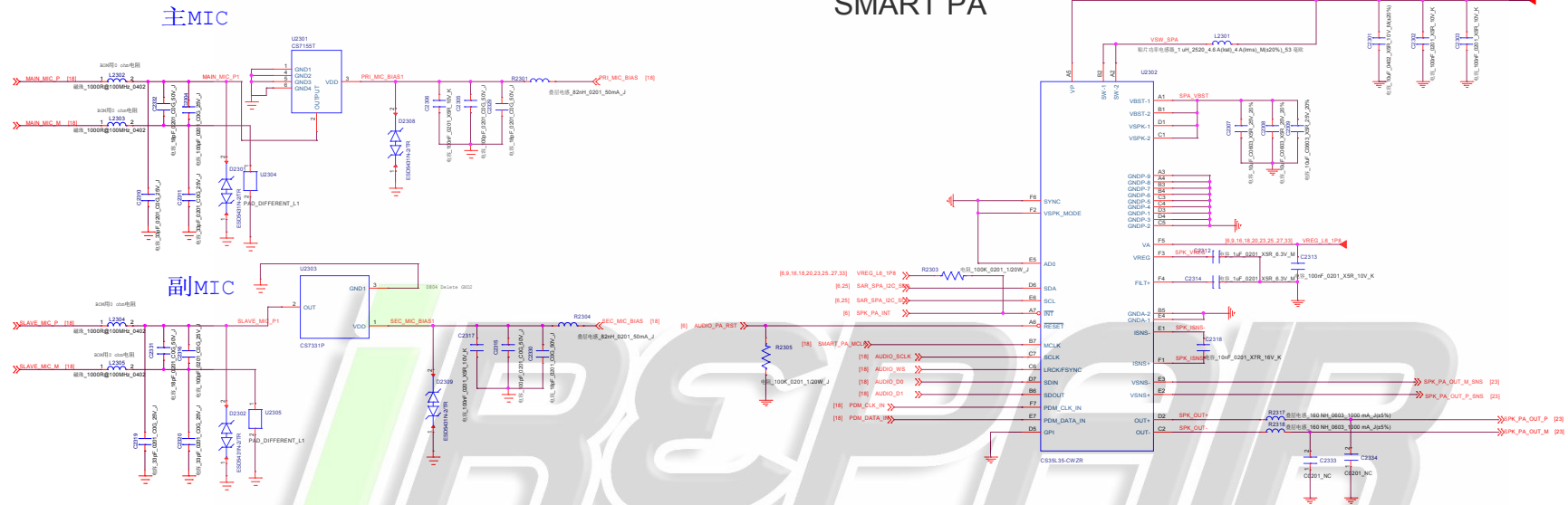


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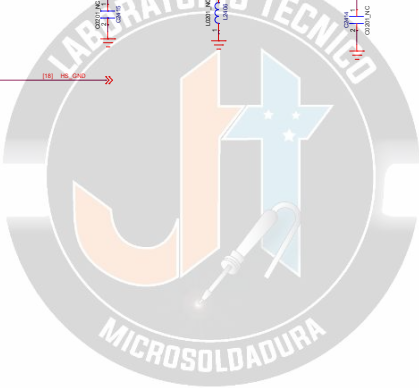
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SMART PA



RCV_MIC_Headphone

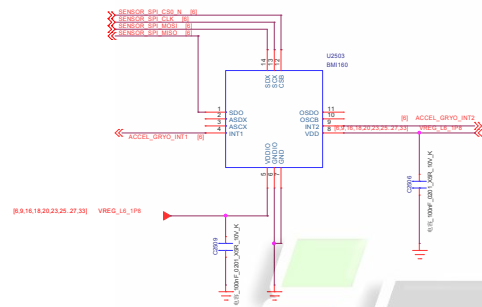
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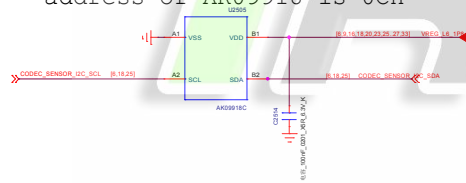
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ACCELEROMETER+GYRO

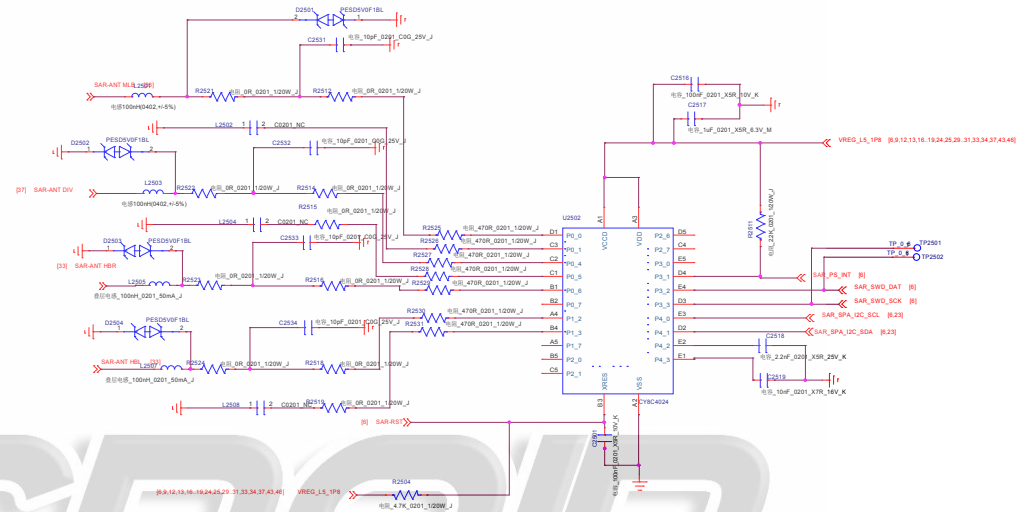
I2C ADDRESS 0X68 0b1101000



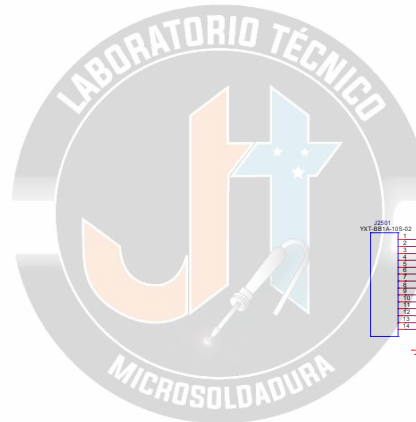
Magnetic Sensor
address of AK09918 is 0Ch



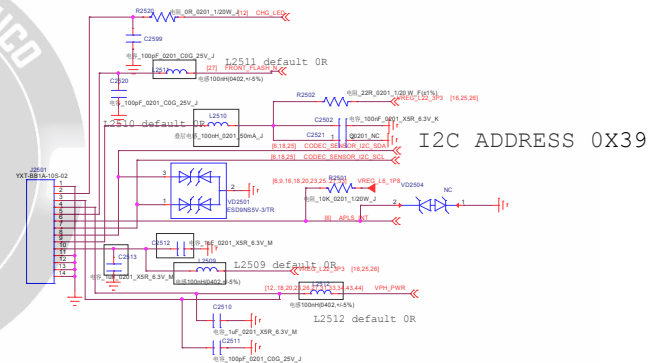
address of AK09918 is 0Ch



ALS/P SENSOR+Front Flash+Charge LED CONN



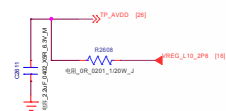
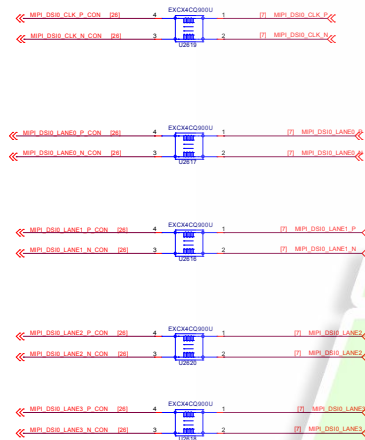
LED change to A-SP194W1D-C01-4T-AR2.4



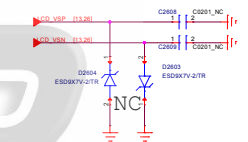
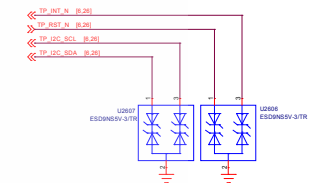
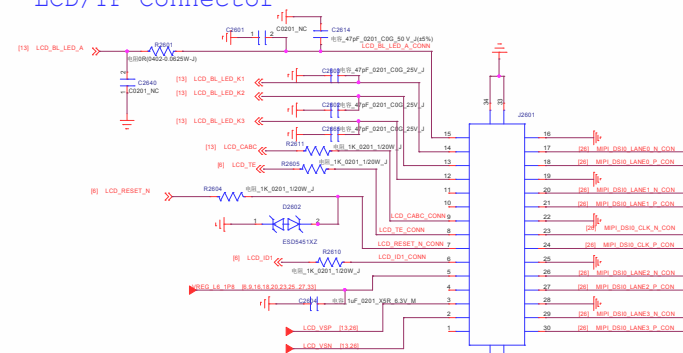
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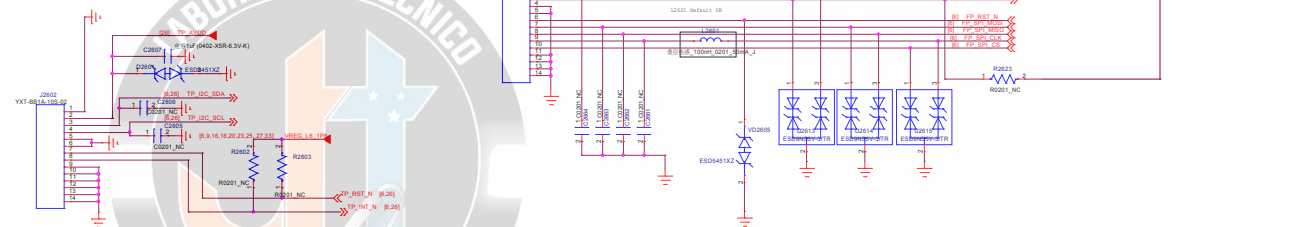
LCD MIPI Common Mode Filter



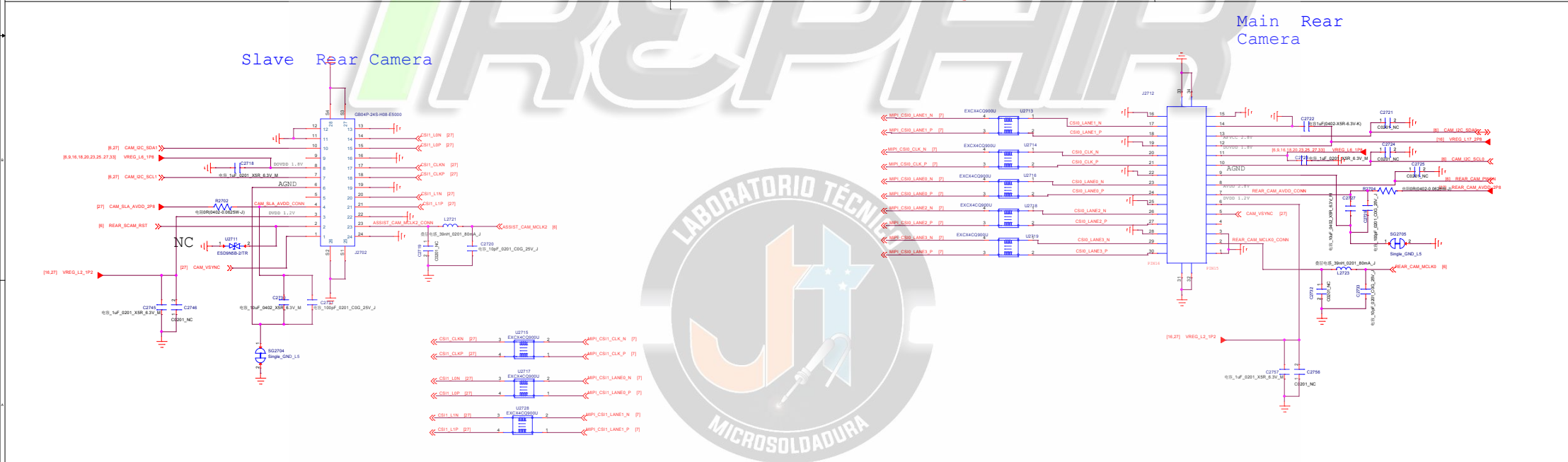
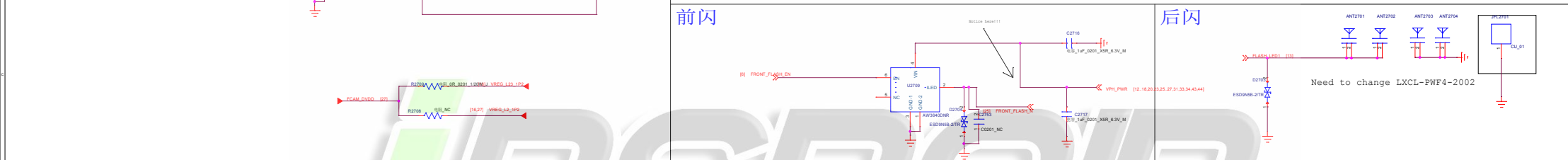
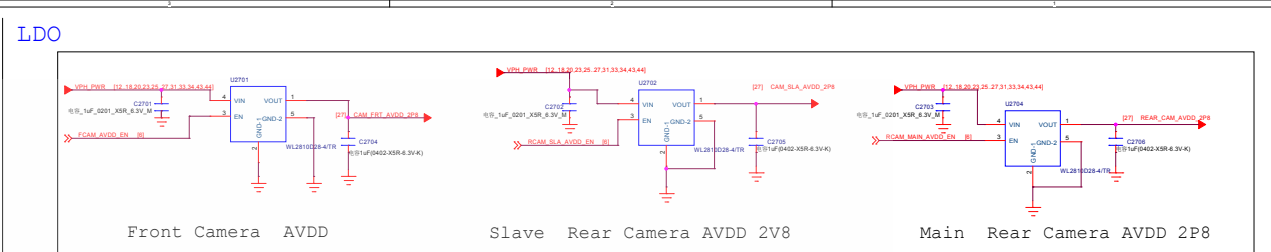
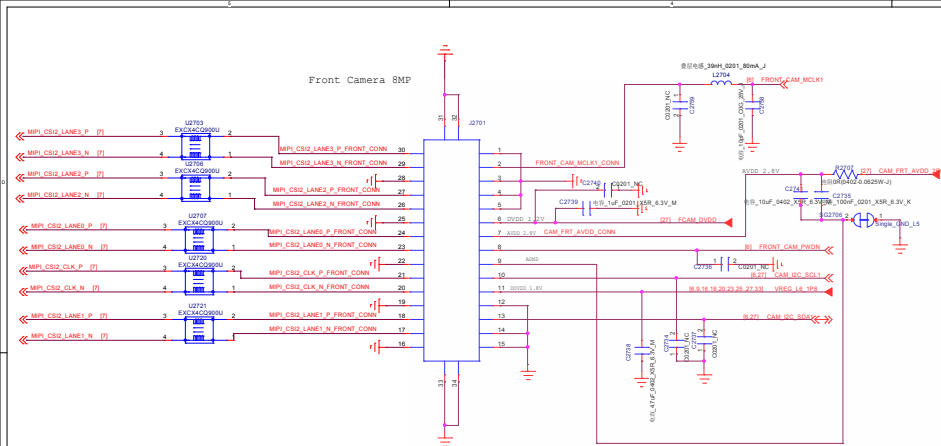
LCD/TP-Connector



FPS-Connector



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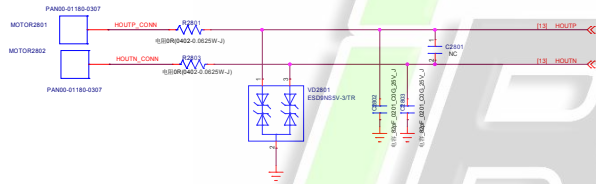


.Main/Slave Camera and Flash

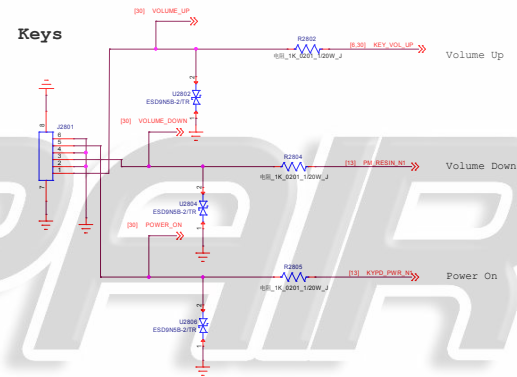
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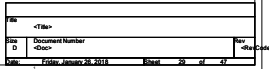
音量键&开关机键

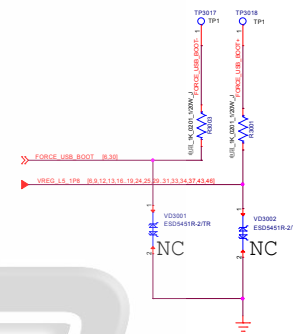
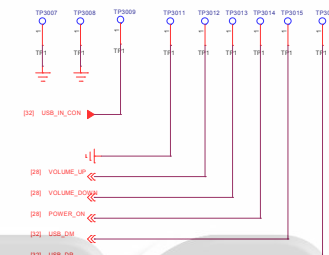
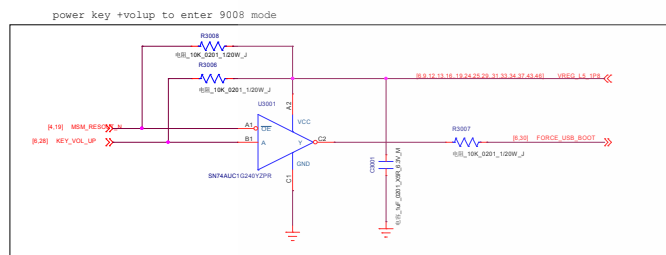
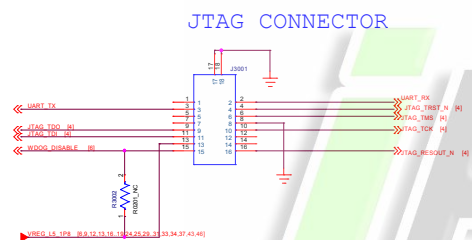
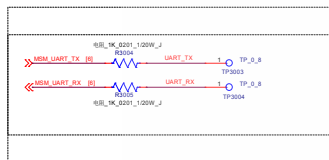
MOTOR



Keys





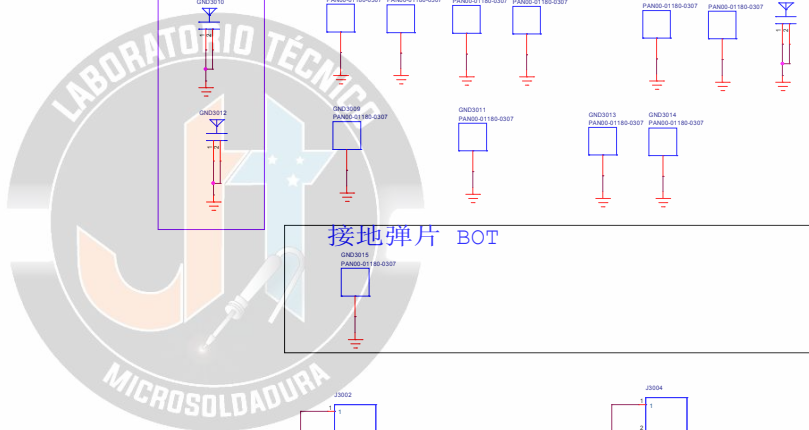


PCBA AUTOMATIC TEST point

接地弹片 TOP

接地弹片 BOT

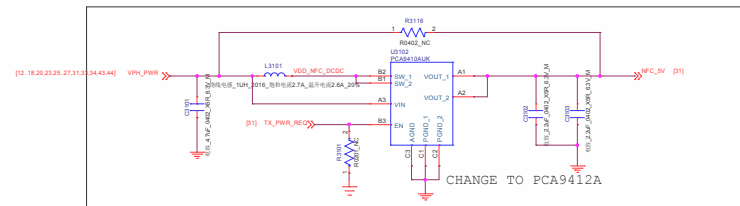
TP/GND/Shielding



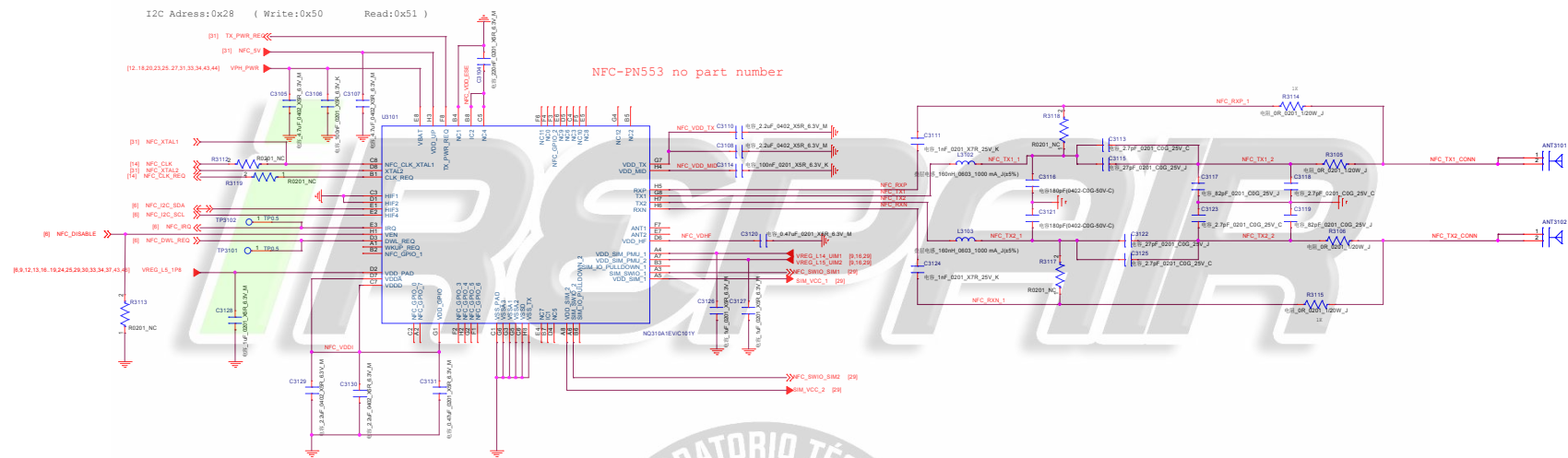
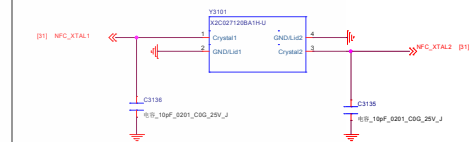
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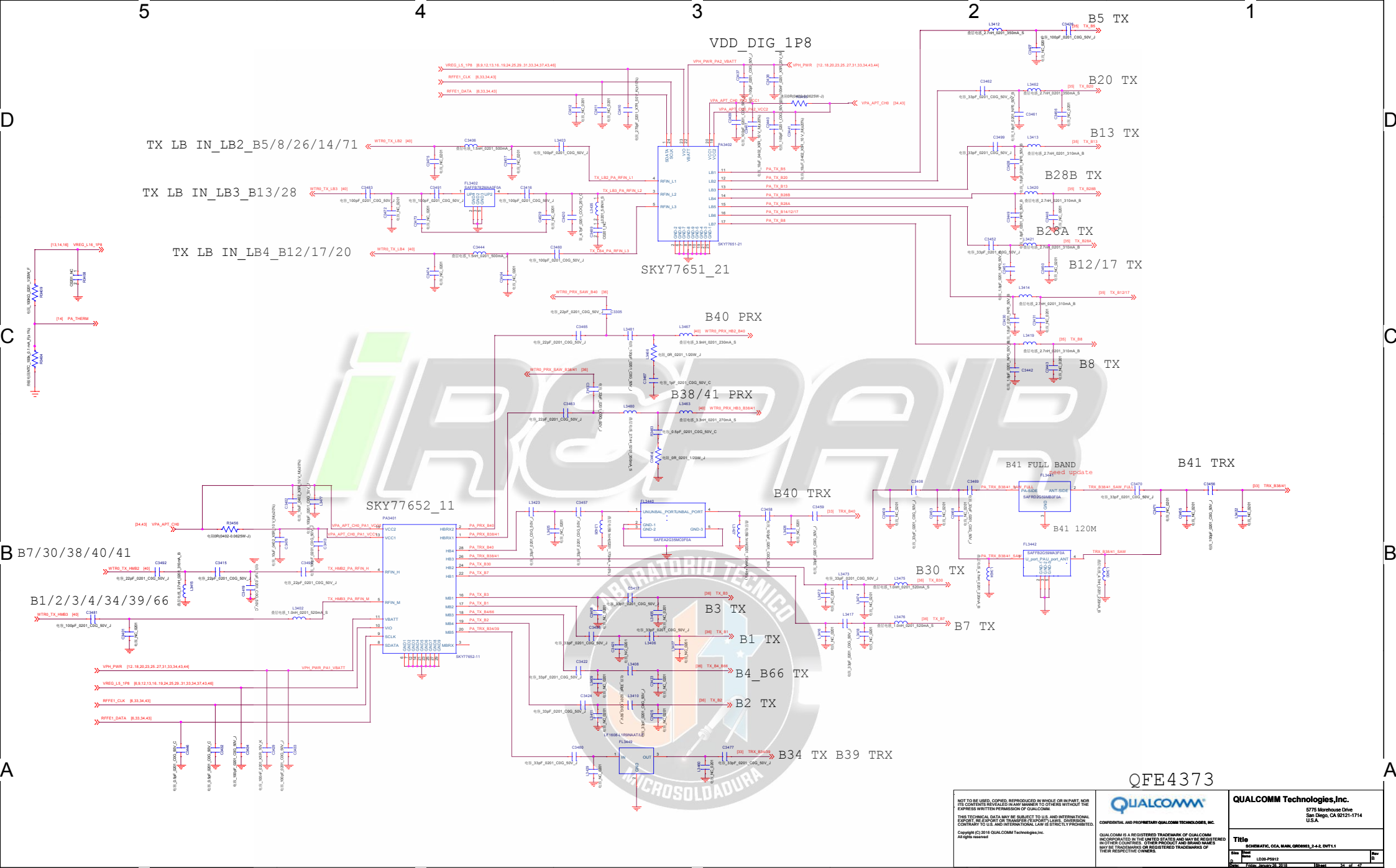
The purpose of this chapter is to propose several application schematics for: NQ310 and NQ330 chips in VF8GA package

DCDC POWER FOR NFC



27.12MHz Crystal





TX B3
TX B1

B1 DPX

B1/B3 QPX

B3 DPX

B2 DPX

B25/66 QPX

B4 PPX

B4:D6DA2G132K2D4
B66:D6DA2G155K2T2

PRX B25

PRX B66

B40 PRx

B30 DRX saw: SAFFB2G35KA1F0A

B30 DPX

B38/41 PRx SAFRD2G59MA0F0A

B7 DPX

change to D6HQ2G65DP01

B34/39 PRx

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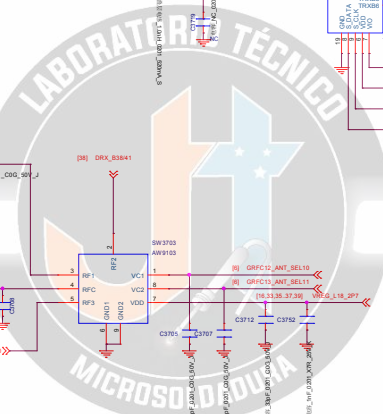
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San Diego, CA 92121-1714
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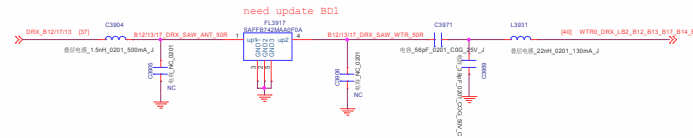
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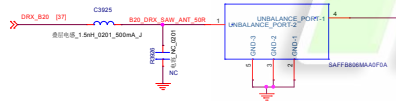


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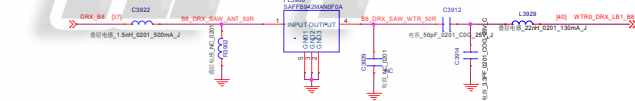
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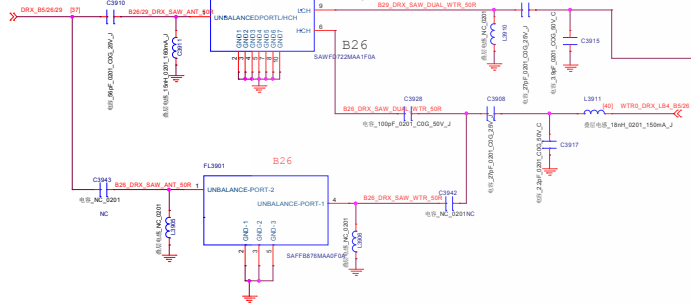
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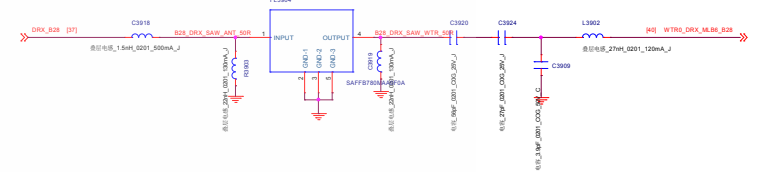
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B5/26/29 DRX

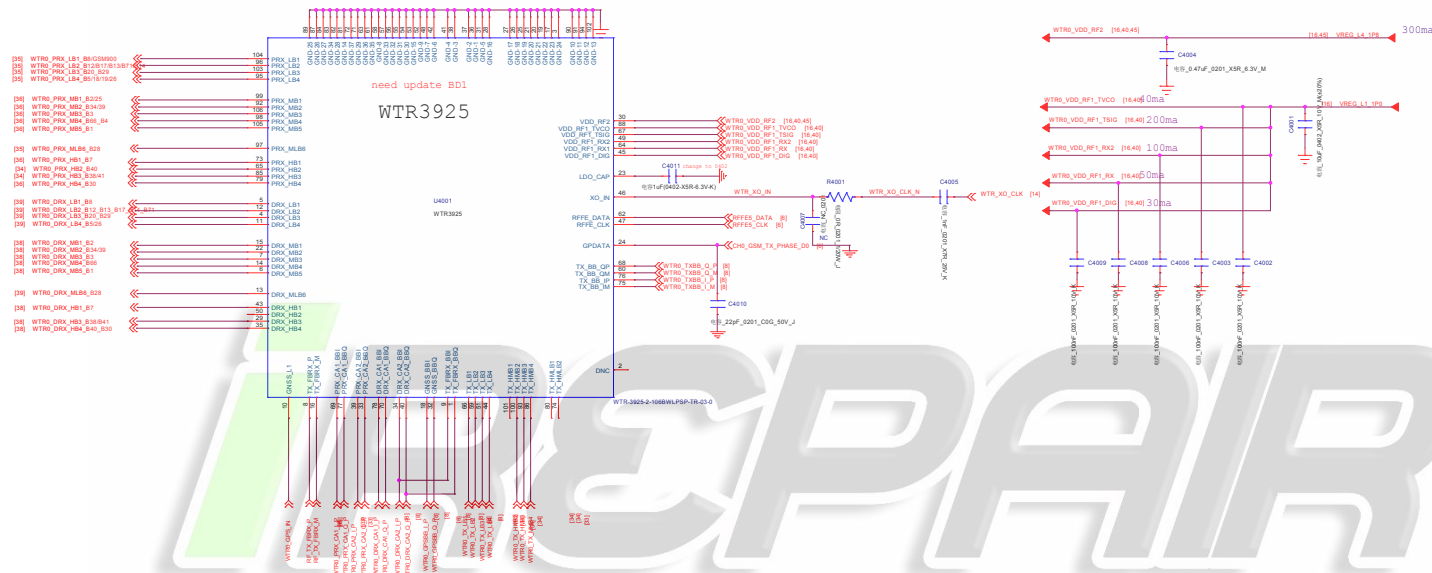


B28 DRX



DRX LB

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WTR 2965_0

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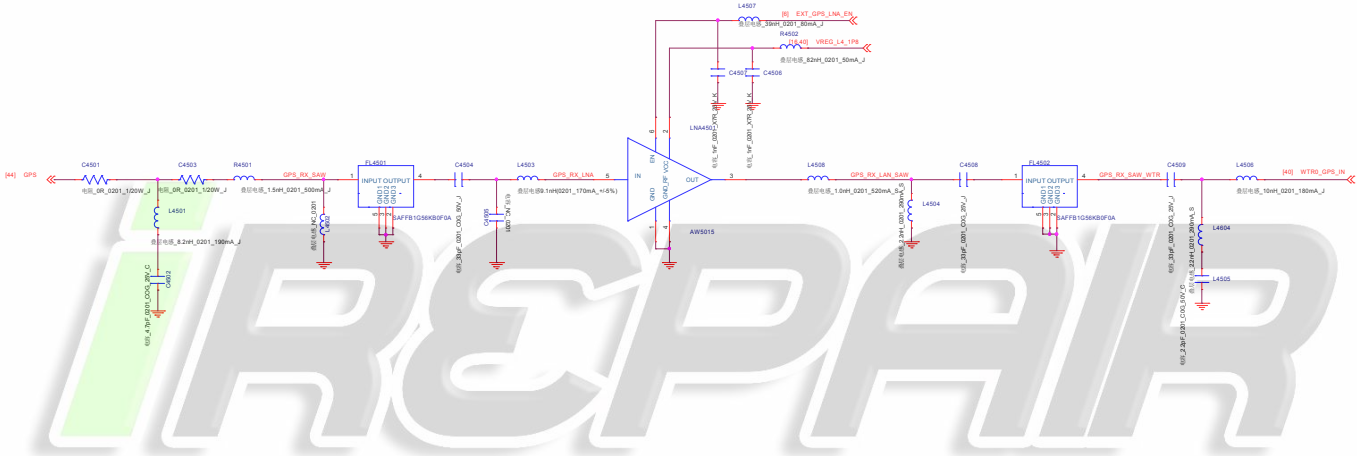
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GPS



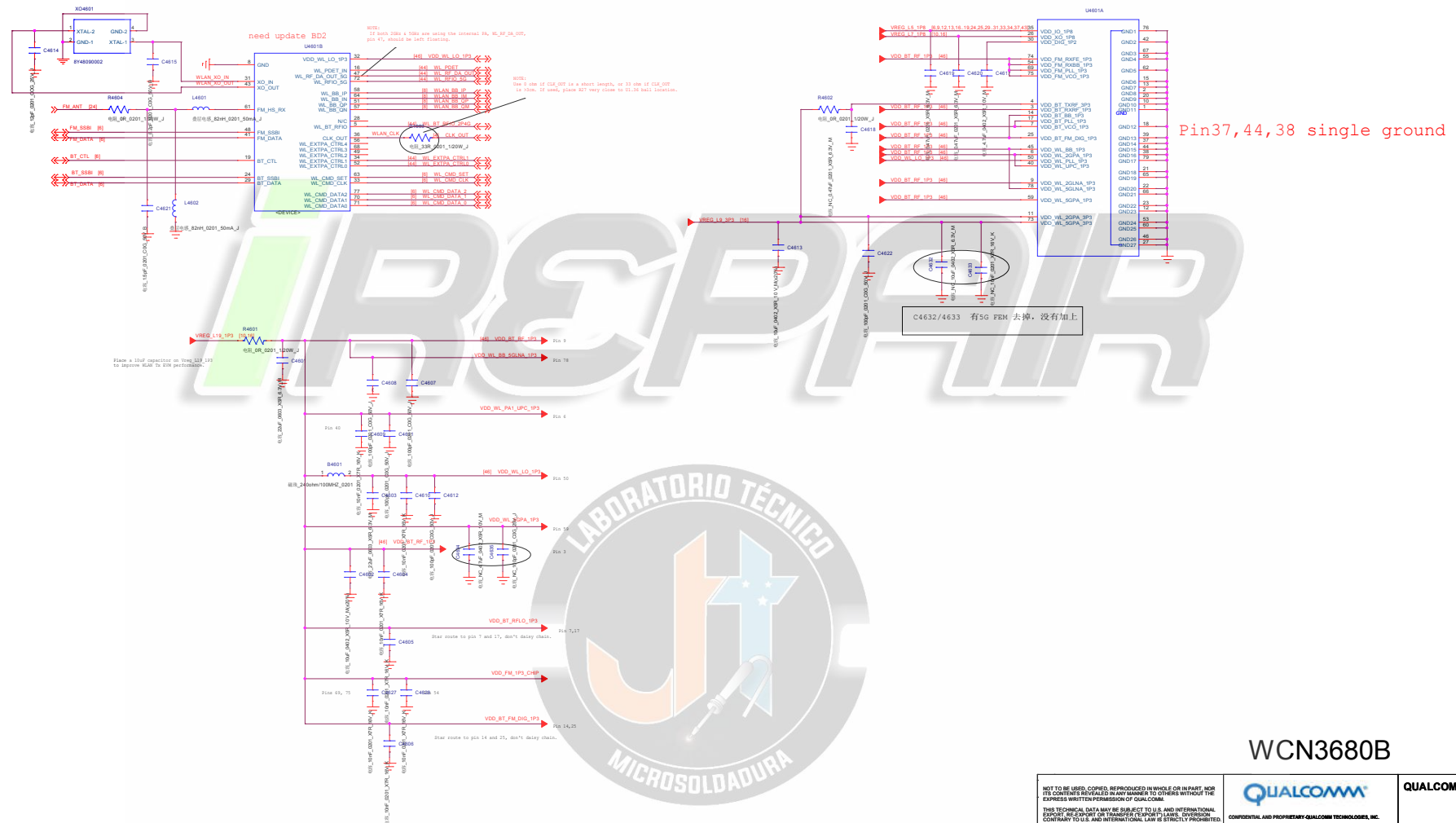
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MSM8953 GPIO Configuration For QRD8953

GPIO_0	NFC_SPI_ESE_MOSI	GPIO_41		GPIO_82	FM_DATA
GPIO_1	NFC_SPI_ESE_MISO	GPIO_42	ACCL_INT1	GPIO_83	BT_CTL
GPIO_2	NFC_SPI_ESE_CS_N	GPIO_43	ALSP_INT_N	GPIO_84	BT_DATA
GPIO_3	NFC_SPI_ESE_CLK	GPIO_44	MAG_DRDY_INT	GPIO_85	KEY_VOL_UP_N
GPIO_4	UART_MSM_TX	GPIO_45	GYRO_INT	GPIO_86	
GPIO_5	UART_MSM_RX	GPIO_46	SCAM_D_ELDO_EN	GPIO_87	
GPIO_6		GPIO_47	SENSOR_SPI_CS2_N	GPIO_88	
GPIO_7		GPIO_48	FP_INT_N	GPIO_89	
GPIO_8		GPIO_49	UIM_BATT_ALARM	GPIO_90	
GPIO_9		GPIO_50	WCD_ELDO_EN	GPIO_91	
GPIO_10	TP_I2C_SDA	GPIO_51	UIM1_DATA	GPIO_92	
GPIO_11	TP_I2C_SCL	GPIO_52	UIM1_CLK	GPIO_93	SCAM_ID
GPIO_12		GPIO_53	UIM1_RESET	GPIO_94	
GPIO_13		GPIO_54		GPIO_95	
GPIO_14	SENSOR_I2C_SDA	GPIO_55	UIM2_DATA	GPIO_96	WSA_EN
GPIO_15	SENSOR_I2C_SCL	GPIO_56	UIM2_CLK	GPIO_97	
GPIO_16	NFC_DISABLE	GPIO_57	UIM2_RESET	GPIO_98	
GPIO_17	NFC_IRQ	GPIO_58		GPIO_99	
GPIO_18	NFC_I2C_SDA	GPIO_59		GPIO_100	GRFC0_SEL
GPIO_19	NFC_I2C_SCL	GPIO_60	GP_PDM_A0	GPIO_101	GRFC1_SEL
GPIO_20	SENSOR_SPI_MOSI	GPIO_61	LCD_RST_N	GPIO_102	GRFC2_SEL
GPIO_21	SENSOR_SPI_MISO	GPIO_62	NFC_DWL_REQ	GPIO_103	GRFC3_SEL
GPIO_22	SENSOR_SPI_CS0_N	GPIO_63		GPIO_104	GRFC4_SEL
GPIO_23	SENSOR_SPI_CLK	GPIO_64	TP_RST_N	GPIO_105	GRFC5_SEL
GPIO_24	LCD_TE0	GPIO_65	TP_INT_N	GPIO_106	WDOG_DISABLE
GPIO_25		GPIO_66		GPIO_107	GRFC7_SEL
GPIO_26	CAM_MCLK0	GPIO_67	WCD_RST_N	GPIO_108	GRFC8_SEL
GPIO_27	CAM_MCLK1	GPIO_68		GPIO_109	GRFC9_SEL
GPIO_28		GPIO_69	WCD_MSM_MCLK	GPIO_110	
GPIO_29	CAM_I2C_SDA0	GPIO_70	SLIMBUS_CLK	GPIO_111	
GPIO_30	CAM_I2C_SCL0	GPIO_71	SLIMBUS_DATA0	GPIO_112	
GPIO_31	CAM_I2C_SDA1	GPIO_72		GPIO_113	
GPIO_32	CAM_I2C_SDA1	GPIO_73	WCD_INTR1	GPIO_114	
GPIO_33		GPIO_74	WCD_INTR2	GPIO_115	
GPIO_34	FLASH_STROBE_NOW	GPIO_75	BT_SSBI	GPIO_116	EXT_GPS_LNA_EN
GPIO_35		GPIO_76	WL_CMD_DATA_2	GPIO_117	CH0_GSM_TX_PHASE_D0
GPIO_36		GPIO_77	WL_CMD_DATA_1	GPIO_118	RFEE1_CLK
GPIO_37	FORCE_USB_BOOT	GPIO_78	WL_CMD_DATA_0	GPIO_119	RFEE1_DATA
GPIO_38		GPIO_79	WL_CMD_SET	GPIO_120	RFEE2_CLK
GPIO_39	MCAM_PWD_N	GPIO_80	WL_CMD_CLK	GPIO_121	RFEE2_DATA
GPIO_40	MCAM_RST_N	GPIO_81	FM_SSBI	GPIO_122	RFEE4_CLK

GPIO_123	RFEE4_DATA	GPIO_134	
GPIO_124		GPIO_135	FP_SPI_CLK
GPIO_125		GPIO_136	FP_SPI_CS
GPIO_126	RFEE3_CLK	GPIO_137	FP_SPI_MOSI
GPIO_127	RFEE3_DATA	GPIO_138	FP_SPI_MISO
GPIO_128		GPIO_139	USB_SS_SEL
GPIO_129	SCAM_RST_N	GPIO_140	FP_RST_N
GPIO_130	SCAM_PWD_N	GPIO_141	NFC_ESE_PWR_REQ
GPIO_131			
GPIO_132			
GPIO_133	SDCARD_DET_N		

PMI8952 GPIO/MPP Configuration For QRD8953

GPIO_1		MPP_1	
GPIO_2		MPP_2	GREEN_LED
		MPP_3	
		MPP_4	FLASH_STROBE_NOW

PM8953 GPIO/MPP Configuration For QRD8953

GPIO_1	CODEC_DIV_CLK	MPP_1	VDD_PX_BIAS_MPP_1
GPIO_2	NFC_CLK_REQ (SDCARD_DET_N)		
GPIO_3	UIM_BATT_ALARM	MPP_3	VREF_DAC_MPP_3
GPIO_4		MPP_4	QUIET_THERM
GPIO_5			
GPIO_6			
GPIO_7	VBUS_USB_IN		
GPIO_8	USB_ID		

GPIO TABLE

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